



CONFIDENTIAL - INTERNAL USE

Clear Build USA

Master Product Document

This is the single source of truth for the Clear Build USA platform - covering product vision, features, architecture, security, database design, APIs, testing, deployment, and launch plan.

PRODUCT	VERSION	DATE	STATUS
Clear Build USA	1.0 - Launch	May 2026	Production-Ready
LIVE URL			
clearbuildusa.com			

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1 Product Overview & Vision

MISSION

Clear Build USA gives US construction contractors a single, professional platform to manage their entire business - from first lead to final payment - replacing scattered spreadsheets, paper invoices, and disconnected apps.

What is Clear Build USA?

Clear Build USA is a cloud-based **construction business management SaaS** built for US-based contractors, general contractors, remodelers, and specialty trades. It covers the complete business lifecycle:

Lead Capture → Quoting → Project Management → Invoicing → Payment Collection

The platform eliminates the need for separate tools like QuickBooks (invoicing), Calendly (scheduling), Slack (communications), and paper/Excel (quotes). Everything lives in one place, accessible on any device.

Core Value Propositions

Value	Description
Professional image	Branded quotes and invoices with online approval - contractors look like established businesses, not sole proprietors with handwritten notes.
Faster payment	Online payment via Stripe, automated invoice reminders, and clear payment tracking reduce days-sales-outstanding.
Less admin time	Auto-numbered documents, change orders linked to projects, and one-click quote-to-invoice conversion cut paperwork by hours per week.
Team coordination	Assign crews to projects, share project visibility with clients, and manage roles so everyone sees only what they need.
Client communication	Unified inbox for Email, SMS, and WhatsApp communications. Customers can approve quotes online - no back-and-forth phone calls.
Business intelligence	Dashboard with revenue trends, outstanding amounts, and pending quotes gives owners an instant business health check.

Product Vision (3-Year)

Become the **#1 all-in-one business platform for US construction contractors with 1–50 employees**.
Expand from core job management into:

- Material procurement and supplier management

- Payroll and labor hour tracking
- AI-driven estimate generation from job photos
- Customer-facing project portal with live progress updates
- Integration marketplace (QuickBooks, Xero, DocuSign, Procore)

Target Geography & Market

Segment	Detail
Geography	United States (all 50 states, primary focus: Sun Belt, Midwest construction markets)
Business size	1–50 employees; sweet spot is 2–15 employees
Trades	General Contractor, Remodeler, Electrician, Plumber, HVAC, Painter, Landscaper, Roofer, Flooring
TAM	~3.8 million US construction establishments (US Census); ~700K actively seeking digital tools
Pricing anchor	\$49/month - positioned well below enterprise tools (\$200+/mo) and above free spreadsheets

2 Business Goals & Success Metrics

Launch Goals (First 90 Days)

Goal	Target	Metric
Paying subscribers	50 businesses	Stripe active subscriptions
Trial sign-ups	300 free trials	businesses.trial_ends_at set
Trial → Paid conversion	≥ 15%	Trials that upgraded within 15 days
Monthly Recurring Revenue	\$2,450 MRR	Stripe MRR dashboard
Churn rate	< 5% / month	Stripe cancellations
Support tickets	< 2 per 10 users	support_tickets table
NPS score	≥ 40	In-app feedback survey

6-Month Goals

Goal	Target
Paying subscribers	250
MRR	\$12,500
Feature completeness	Project Gantt chart, mobile app, QuickBooks export
Reviews on G2 / Capterra	25+ reviews, 4.5★ average

Key Performance Indicators

Activation Rate

% of trial users who create their first quote within 7 days. Target: 60%+

Retention (Day 30)

% of paid users still active at 30 days. Target: 85%+

ARPU

Average Revenue Per User. Target: \$49–\$55 blended (monthly vs annual mix)

Time to First Value

Minutes from sign-up to first quote created. Target: < 10 minutes

3 User Personas

MR

Mike R. - Solo Contractor

General Contractor · 1 person · Texas

Runs his own GC business, does everything himself. Currently uses paper estimates and QuickBooks. Frustrated by chasing payments and looking unprofessional to clients.

Needs: Fast quoting, online payment, professional appearance

Pain: No time for complex software. Must be simple.

JS

Jennifer S. - Small GC Owner

GC Owner · 8 employees · Florida

Manages a crew of 8. Runs multiple projects simultaneously. Uses Excel for job tracking, Google Calendar for scheduling, and texts crew assignments. Overwhelmed by admin.

Needs: Multi-project dashboard, team roles, change orders

Pain: No single view of all jobs + finances

KP

Kevin P. - Project Manager

Project Manager · 15-person GC · Ohio

Assigned to manage 4–6 active projects. Needs to track tasks, communicate with clients, and log daily updates without going through the owner for every action.

Needs: Project tasks, team view, client communication log

Pain: Owner bottleneck; no visibility sharing with clients

AL

Amy L. - Office Administrator

Office Admin · 12-person Remodeler · California

Handles all invoicing, payment follow-ups, and customer communication. Needs a clean interface with no surprises, and confidence that what she sends looks professional.

Needs: Invoice management, email templates, payment tracking

Pain: Manual reminder emails; no payment status visibility

Role Permissions Overview

Capability	Owner	Admin	Staff	Crew
View dashboard & contacts	✓	✓	✓	✓
Create/edit quotes & invoices	✓	✓	✓	✗
Record payments	✓	✓	✗	✗
Approve change orders	✓	✓	✗	✗
Send notifications/emails	✓	✓	✗	✗
Manage team members	✓	✓	✗	✗
Access billing & settings	✓	✗	✗	✗
View assigned projects	✓	✓	✓	✓

4.1 Authentication & User Management

- Email + password registration with mandatory email verification
- Secure login with JWT session (15-day expiry, stored in HTTP-only cookie)
- Forgot password / reset password via email token (1-hour expiry)
- Change password from settings (requires current password)
- Multi-business support - one user can belong to multiple businesses
- Business switcher in the top bar to switch active business context
- Role-based access control: Owner, Admin, Staff, Crew
- Immediate session revocation when a team member is removed
- Resend verification email for unverified accounts

4.2 Dashboard

- Personalised greeting (Good Morning/Afternoon/Evening, [First Name])
- 4 live stat cards: Outstanding Invoices Amount, Pending Quotes, Active Projects, Payments Received (MTD)
- 6-month revenue bar chart with month-by-month breakdown
- Recent activity feed (latest quotes, invoices, payments, contacts)
- Outstanding invoices list with quick "View" links
- Quick action buttons: New Contact, New Quote, New Invoice, Payments
- "No activity yet" state for new businesses

4.3 Contact Management

- Unified contacts list with search and type filter
- Three contact types: **Lead**, **Customer**, **Direct Contact**
- Lead status pipeline: New Lead → In Conversation → Meeting Scheduled → Site Visit → Proposal Sent → Negotiation → Won / Lost
- Heat scoring: Hot (New Lead / In Conversation), Warm (Meeting/Site/Proposal/Negotiation), Cold (Lost/other)
- Contact detail page with full activity timeline
- Lifetime value (total invoiced) per contact
- Separate Leads and Customers views with dedicated filters
- Convert Lead → Customer with one click
- Leads page: 4 stat cards (Total, Hot, Warm, Cold), Heat badges
- Customers page: Grid/List toggle, star ratings, avatar cards
- Server-side pagination (50 records per page) on all list views

4.4 Quotes

- Create quotes with multiple line items (description, quantity, unit price, discount, tax)
- Auto-numbering with configurable prefix (e.g., Q-0001)
- 3-column metadata: Customer, Job Site, Valid Through (default: today +30 days)
- Send channel toggles: Email, SMS, WhatsApp
- Activity sidebar showing all actions taken on the quote
- Quote status flow: Draft → Sent → Approved / Rejected
- Public quote preview page (no login required) - customer can Approve or Request Changes
- Only "Sent" quotes can be approved (prevents double-approval exploit)
- Convert approved quote to invoice with one click
- 5 stat cards on list: Total, Draft, Sent, Approved, Rejected
- Filter tabs by status; avatar initials for quick contact identification

4.5 Invoices

- Create invoices with line items (same structure as quotes)
- Auto-numbering with configurable prefix (e.g., INV-0001)
- Payment options panel: due date, deposit %, payment method
- Schedule reminders: days before due, communication channel
- Internal notes visible only to your team
- Invoice status flow: Draft → Sent → Due → Overdue → Paid / Partial
- Record payment modal: Full / 50% / Deposit / Custom amount tabs
- Payment methods: Card, Bank Transfer, Check, Cash
- Online payment via Stripe (customer clicks "Pay \$X" in invoice)
- Payment summary sidebar with progress bar on invoice detail
- Activity timeline on invoice detail
- Automatic project creation when no project is selected on invoice creation
- 4 stat cards on list: Total, Unpaid, Overdue, Paid (MTD)

4.6 Projects

- Project list with gradient colour cards and progress bars
- "Day X of Y" timeline display based on start/end dates
- Budget tracking: budget amount, deposit %, financial summary
- Team assignment: assign specific business members to a project
- Task management: create tasks, assign due dates, mark complete
- Client visibility toggle: share or hide project details from the customer
- Project status: Planning, In Progress, On Hold, Completed, Cancelled
- Link invoices, quotes, and change orders to projects
- Auto-numbering with configurable prefix (e.g., PRJ-0001)
- Project updates log for daily notes and progress photos

4.7 Payments

- Full payment history across all invoices

- 4 stat cards: Card, Bank, Check, Cash totals
- Mini bar chart of payments over time
- Payment reversal (Owner/Admin only)
- Stripe webhook integration for automatic payment status updates
- "Cleared" badge for verified digital payments

4.8 Change Orders

- Create change orders linked to existing projects
- Line items with description, quantity, unit price
- Status: Draft → Sent → Approved / Rejected / Void
- Role-guarded: only Owner/Admin can approve or void
- 3 stat cards: Draft, Pending Approval, Approved

4.9 Scheduling

- **Availability Slots:** Create individual time slots with date, start/end time, meeting type, location, timezone. Support recurring slots (daily/weekly/monthly).
- **Booking Links:** Generate shareable booking link tokens tied to specific slots. Include message to recipient, expiry date.
- **Public Booking Page:** Customer-facing page (no login required) to select a slot and confirm a meeting.
- **Meeting Management:** Track all meetings with statuses: Scheduled, Confirmed, Completed, Canceled, No Show, Rescheduled
- Meeting types: Consultation, Site Visit, Estimate Meeting, Project Discussion, Final Walkthrough, Follow-up Call, Other

4.10 Communications

- Two-panel chat-style inbox grouped by contact
- Channel support: Email, SMS, WhatsApp, Phone Call, Note
- Filter tabs by channel (All, Email, SMS, WhatsApp, Note)
- Search contacts and message content
- Log manual communication records (call notes, manual emails)
- Mobile-responsive: switches between list and chat views

4.11 Team Management

- Invite team members by email (sends invitation email with token link)
- 4 roles: Owner, Admin, Staff, Crew - each with distinct permissions
- 4 stat cards on team page: Total, Online, Admins, Crew
- Roles & Permissions reference table on team page
- Remove team member - immediately invalidates their JWT (force_logout_at)
- Invited users who don't yet have an account get prompted to register first

4.12 Templates

- Reusable message templates for notifications, quotes, invoices

- Create, edit, delete, and duplicate templates
- Variable support: {{contact_name}}, {{business_name}}, etc.

4.13 Notifications

- In-app notification feed
- Send notifications via Email or WhatsApp to contacts (Owner/Admin only)
- Default notification templates: Quote Sent, Invoice Due Reminder, Payment Received
- Mark as read, mark all as read

4.14 Settings (9 Sections)

- **Business Profile:** Name, type, address, phone, logo upload
- **Preferences:** Default currency, date format, fiscal year start
- **Tax & Numbering:** Default tax rate, quote/invoice/project number prefixes
- **Notification Preferences:** Toggle email/push for various events
- **Email Templates:** Customise outgoing email body/subject per event type
- **Integrations:** Stripe connection status, webhook endpoint configuration
- **Region & Language:** Timezone, language preference
- **Security:** Change password (requires current password)
- **API & Webhooks:** Configure outbound webhooks for 8 event types

4.15 AI Chatbot

- Powered by Google Gemini 2.0 Flash
- Business assistant accessible from any page
- Contextual help for using the platform
- Graceful error handling - returns user-friendly messages if AI is unavailable

4.16 Help & Support

- FAQ system with search
- Support ticket creation and management
- Ticket status: Open, In Progress, Resolved, Closed
- Message thread within each ticket

4.17 Profile Management

- User profile: bio, skill tags, recent work grid
- Photo upload
- Change display name, bio, and social links
- Danger zone: delete account

4.18 Admin Panel (Internal)

- Separate admin login at /admin
- Dashboard: MRR, total users, active businesses, churn analytics

- User management: list, view, edit, deactivate users; impersonation
- Business management: view all businesses, their subscription status
- Subscription management: view, upgrade, cancel, override subscriptions
- Plan management: create/edit plans and pricing
- Broadcast messages: send announcements to all or filtered users
- Audit logs: full platform activity log with filters
- Stripe sync: reconcile Stripe data with database

5 Complete Screen List & User Flows

5.1 All Application Screens

#	Screen Name	Route	Auth Required	Status
01	Landing Page	/	No	Live
02	Login	/login	No	Live
03	Register	/register	No	Live
04	Email Verification	/verify-email	No	Live
05	Forgot Password	/forgot-password	No	Live
06	Reset Password	/reset-password	No	Live
07	Business Setup	/business-setup	Yes (no business)	Live
08	Dashboard	/dashboard	Yes	Live
09	Contacts List	/contacts	Yes	Live
10	Contact Detail	/contacts/[id]	Yes	Live
11	New Contact	/contacts/new	Yes	Live
12	Leads List	/leads	Yes	Live
13	Customers List	/customers	Yes	Live
14	Quotes List	/quotes	Yes	Live
15	Create Quote	/quotes/new	Yes	Live
16	Quote Detail	/quotes/[id]	Yes	Live
17	Quote Preview (Public)	/quotes/[id]/preview	No	Live
18	Invoices List	/invoices	Yes	Live
19	Create Invoice	/invoices/new	Yes	Live
20	Invoice Detail	/invoices/[id]	Yes	Live

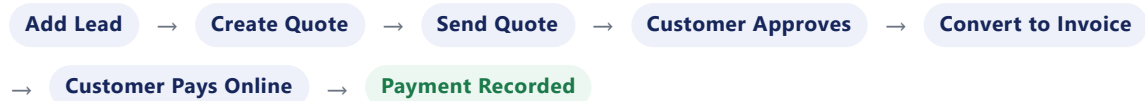
21	Projects List	/projects	Yes	Live
22	Create Project	/projects/new	Yes	Live
23	Project Detail	/projects/[id]	Yes	Live
24	Payments List	/payments	Yes	Live
25	Change Orders List	/change-orders	Yes	Live
26	Create Change Order	/change-orders/new	Yes	Live
27	Change Order Detail	/change-orders/[id]	Yes	Live
28	Scheduling	/scheduling	Yes	Live
29	Public Booking Page	/book/[token]	No	Live
30	Communications	/communications	Yes	Live
31	Item Requirements	/item-requirements	Yes	Live
32	Templates	/templates	Yes	Live
33	Notifications	/notifications	Yes	Live
34	Team Management	/team	Yes	Live
35	Help & Support	/help	Yes	Live
36	Settings	/settings	Yes	Live
37	Subscription & Billing	/subscription	Yes	Live
38	Profile	/profile	Yes	Live
39	Profile Edit	/profile/edit	Yes	Live
40	Audit Log	/audit-log	Yes	Live
41	Feedback	/feedback	Yes	Live
42	Project Updates	/project-updates	Yes	Live

5.2 Core User Flows

Flow A: New Contractor Onboarding



Flow B: Lead to Paid Invoice



Flow C: Project Execution



Flow D: Team Member Invitation



Flow E: Customer Books a Meeting



6 Subscription Plans & Pricing

Plan	Price	Billing	Features	Stripe Plan ID
Free Trial	\$0	15 days, no card required	All features included, unlimited contacts, quotes & invoices, team access, scheduling, email support	trial (internal)
Monthly Pro	\$49/month	Monthly, cancel any time	Everything in Trial + Unlimited projects, scheduling & bookings, team management, templates library, priority support	pro_monthly
Yearly Pro	\$490/year	Annual (saves \$98 = 2 months free)	Everything in Monthly + Annual billing receipt, early access to new features	pro_yearly

Billing Flow

1. User clicks "Start Monthly Plan" or "Start Yearly Plan" on /subscription
2. App calls POST /api/billing/checkout → creates Stripe Checkout Session
3. User redirected to Stripe-hosted checkout page

4. On success, Stripe sends webhook to POST /api/webhooks/stripe
5. Webhook updates subscriptions table with Stripe subscription ID, status, and plan
6. User redirected to /subscription?success=1 with confirmation toast

Trial Expiry Behaviour

- After 15 days, `restricted_mode = true` is set on the business
- Restricted mode shows upgrade prompts on create flows
- Read-only access to existing data remains available during restriction

7 Technical Architecture

7.1 Technology Stack

Layer	Technology	Version	Purpose
Frontend Framework	Next.js (App Router)	15	Full-stack React framework, SSR + client components
UI Library	React	19	Component model, concurrent features
Language	TypeScript	5.x	Type safety across frontend and API routes
Styling	Tailwind CSS	3.x	Utility-first CSS, custom design system
Database	Supabase (PostgreSQL)	15	Relational database, real-time subscriptions
Auth	Custom JWT (jose + bcryptjs)	-	Signed JWT sessions, password hashing
Email	Resend	v1	Transactional emails (verification, invoices, invites)
Payments	Stripe	v14	Subscription billing, online invoice payments
AI	Google Gemini	2.0 Flash	In-app chatbot assistant
Icons	Lucide React	-	Consistent icon set throughout the app
Deployment	Hostinger	-	Node.js hosting, auto-deploy from GitHub
Version Control	GitHub	-	Source of truth: yogesh-112/cbtermi, branch: master

7.2 Application Architecture

PATTERN: CLIENT PAGES + API ROUTES

All `app/(app)/` pages are "use client" components that call their corresponding `app/api/` routes. The `app/(app)/layout.tsx` is the only server component - it reads the session and injects user, businesses, and `currentBusiness` into the sidebar. This keeps auth server-side while keeping UI interactive.

Authentication Flow

1. User logs in → POST `/api/auth/login` verifies password, signs JWT with jose
2. JWT stored as `cb_session` cookie (HTTP-only, SameSite=Lax)
3. JWT payload: `{ id, name, email, businessId, role }`
4. Every API route calls `requireSession()` from `lib/auth.ts`
5. `middleware.ts` runs before every page render to guard protected routes
6. Unauthenticated → redirect to `/login`; no `businessId` → redirect to `/business-setup`

Multi-Tenancy Pattern

Every API query includes `.eq("business_id", session.businessId)`. The Supabase service role key is used server-side only, bypassing RLS. Application code enforces all tenancy boundaries - no user can see another business's data.

File Structure

Path	Contents
<code>app/(app)/</code>	All authenticated app pages (client components)
<code>app/api/</code>	All API route handlers (server-side)
<code>app/admin/</code>	Admin panel pages
<code>components/</code>	Shared UI components (<code>ui.tsx</code> , <code>layout/Topbar</code> , <code>Sidebar</code>)
<code>lib/</code>	Server utilities: <code>session.ts</code> , <code>auth.ts</code> , <code>supabase.ts</code> , <code>email.ts</code> , <code>utils.ts</code>
<code>app/globals.css</code>	Design system: all custom Tailwind component classes
<code>tailwind.config.js</code>	Brand colors, custom screen breakpoints

8 API Reference

AUTHENTICATION

All private API routes require a valid `cb_session` JWT cookie. Public routes (quote preview, booking pages) are explicitly unauthenticated. All private routes filter data by `session.businessId`.

Authentication Routes

Method	Route	Description
POST	/api/auth/register	Create user account, send verification email
POST	/api/auth/login	Verify credentials, issue JWT cookie
POST	/api/auth/logout	Clear session cookie
GET	/api/auth/me	Return current session user
POST	/api/auth/verify-email	Verify email with token
POST	/api/auth/resend-verification	Resend verification email
POST	/api/auth/forgot-password	Send password reset email
POST	/api/auth/reset-password	Reset password with valid token
POST	/api/auth/change-password	Change password (requires current password)

Core Business Routes

Method	Route	Description
GET/POST	/api/contacts	List (paginated) / Create contact
GET/PATCH/DELETE	/api/contacts/[id]	Get / Update / Delete contact
GET/POST	/api/quotes	List / Create quote
GET/PATCH/DELETE	/api/quotes/[id]	Get / Update / Delete quote
GET/PATCH	/api/quotes/[id]/preview	Public quote view / Customer approval
GET/POST	/api/invoices	List / Create invoice

GET/PATCH/DELETE	/api/invoices/[id]	Get / Update / Delete invoice
POST	/api/invoices/[id]/pay	Initiate Stripe payment for invoice
GET/POST	/api/projects	List / Create project
GET/PATCH/DELETE	/api/projects/[id]	Get / Update / Delete project
GET/POST/PATCH/DELETE	/api/projects/[id]/tasks	Project task CRUD
GET/POST/DELETE	/api/projects/[id]/team	Project team assignment
GET/POST	/api/payments	List / Record payment
POST	/api/payments/[id]/reverse	Reverse a payment (Owner/Admin)
GET/POST	/api/change-orders	List / Create change order
GET/PATCH/DELETE	/api/change-orders/[id]	Get / Update (approve/reject) / Delete

Scheduling Routes

Method	Route	Description
GET/POST	/api/scheduling/slots	List / Create availability slot
PATCH/DELETE	/api/scheduling/slots/[id]	Update / Delete slot
GET/POST	/api/scheduling/booking-links	List / Create booking link
PATCH/DELETE	/api/scheduling/booking-links/[id]	Update / Delete booking link
GET/POST	/api/scheduling/meetings	List / Create meeting
PATCH	/api/scheduling/meetings/[id]	Update meeting status
GET/POST	/api/public/booking/[token]	Public booking page data / Confirm booking

Support & Misc Routes

Method	Route	Description
GET/POST	/api/communications	Communication log list / Send message
GET/POST	/api/notifications	List notifications / Send notification

GET/POST/DELETE	/api/team	List team / Invite member / Remove member
GET/PATCH	/api/settings	Get / Update business settings (whitelisted fields)
GET/POST/PATCH/DELETE	/api/templates	Template CRUD
GET	/api/help/faqs	Search FAQs (max 200-char query)
GET/POST	/api/support/tickets	List / Create support ticket
GET/POST	/api/support/tickets/[id]/messages	Ticket message thread
GET	/api/dashboard	Dashboard stats, chart data, recent activity
GET	/api/search	Global search across contacts, quotes, invoices
POST	/api/upload	File / logo upload
GET/PATCH	/api/profile	Get / Update user profile
GET/PATCH	/api/subscription	Get subscription status
POST	/api/billing/checkout	Create Stripe Checkout session
POST	/api/billing/portal	Open Stripe customer billing portal
POST	/api/webhooks/stripe	Receive Stripe payment webhooks
POST	/api/chatbot	AI chatbot (Gemini 2.0 Flash)
GET/POST	/api/businesses	List user's businesses / Create new business
POST	/api/businesses/switch	Switch active business (re-signs JWT)

Webhook Events (Outbound)

Configurable in Settings → API & Webhooks. Events sent as POST to your configured URL:

invoice.created

invoice.paid

invoice.overdue

quote.approved

quote.rejected

payment.recorded

contact.created

project.updated

9 Database Structure

DATABASE PROVIDER

PostgreSQL hosted on Supabase. The service role key is used server-side (bypasses Row Level Security). All tenancy isolation is enforced in application code via `business_id` filters.

Core Tables

Table	Key Columns	Purpose
users	id, email, password_hash, full_name, email_verified, last_business_id, force_logout_at	All platform users
businesses	id, name, type, owner_id, quote_prefix, invoice_prefix, project_prefix, trial_ends_at, restricted_mode, stripe_customer_id, logo_url	Contractor businesses (multi-tenant root)
business_members	user_id, business_id, role (owner/admin/staff/crew)	Many-to-many user ↔ business mapping
subscriptions	business_id, stripe_subscription_id, plan_id, status, current_period_end	Stripe subscription state per business
contacts	id, business_id, full_name, email, phone, contact_type, lead_status, lifetime_value, address	Leads, customers, and direct contacts

Documents Tables

Table	Key Columns	Purpose
quotes	id, business_id, contact_id, quote_number, title, status, subtotal, tax_amount, total, valid_until, notes, job_site	Quote headers
quote_items	quote_id, description, quantity, unit_price, discount, tax_rate, total	Quote line items
invoices	id, business_id, contact_id, project_id, invoice_number, status, subtotal, tax_amount, total, amount_paid, amount_due, due_date	Invoice headers
invoice_items	invoice_id, description, quantity, unit_price, discount, tax_rate, total	Invoice line items

change_orders	id, business_id, project_id, co_number, status, subtotal, total	Change order headers
change_order_items	change_order_id, description, quantity, unit_price	Change order line items

Project & Operations Tables

Table	Key Columns	Purpose
projects	id, business_id, contact_id, project_number, name, status, budget, deposit_pct, start_date, end_date, client_visible	Project records
project_tasks	project_id, title, description, status, due_date, assigned_to	Tasks within a project
project_team_members	project_id, user_id, role	Team assigned to a project
project_updates	project_id, author_id, content, created_at	Daily log / progress notes
payments	id, business_id, invoice_id, amount, method, stripe_payment_id, recorded_by, reversed_at	Payment records
item_requirements	project_id, name, quantity, unit, estimated_cost, status	Material requirements per project

Communication & System Tables

Table	Key Columns	Purpose
communications	business_id, contact_id, channel, direction, message, subject, sent_at	Communication log
notifications	business_id, user_id, type, title, body, read, sent_at	In-app notifications
team_invitations	business_id, email, token, role, accepted_at, expires_at	Pending team invitations
scheduling_slots	business_id, slot_date, start_time, end_time, meeting_type, location, status, repeat_option	Availability slots

booking_links	business_id, token, title, slot_ids, contact_id, expiry_date	Shareable booking tokens
meetings	business_id, slot_id, contact_id, status, confirmed_at	Confirmed meetings
templates	business_id, name, type, subject, body, variables	Reusable message templates
support_tickets	user_id, subject, status, priority, created_at	Support tickets
audit_events	business_id, user_id, action, resource, resource_id, metadata, created_at	Audit trail for all major actions
feedback	user_id, business_id, rating, comment, created_at	In-app feedback submissions

10 Third-Party Integrations

Service	Purpose	Env Variables	Notes
Supabase	PostgreSQL database, file storage	SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY	Service role key used server-side only. Row-level security bypassed - app enforces tenancy in code.
Stripe	Subscription billing, online invoice payments	STRIPE_SECRET_KEY, STRIPE_WEBHOOK_SECRET, STRIPE_PUBLISHABLE_KEY	Checkout Sessions for upgrades. Webhooks update subscription status. Invoice pay via Checkout Session.
Resend	Transactional email sending	RESEND_API_KEY	Falls back to console mock if key missing (for local dev). Templates: verification, password reset, team invite.
Google Gemini	AI chatbot assistant	GEMINI_API_KEY	Model: gemini-2.0-flash. Wrapped in try/catch - chatbot

gracefully fails if API
unavailable.

Hostinger	Node.js hosting, domain	PORT (auto-set by host)	Auto-deploys from GitHub master branch. App respects \$PORT env var.
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Environment Variables Reference

Variable	Required	Description
SUPABASE_URL	✔ Required	Supabase project URL (https://xxx.supabase.co)
SUPABASE_SERVICE_ROLE_KEY	✔ Required	Service role key - bypasses RLS. Never expose client-side.
JWT_SECRET	✔ Required	Minimum 32-char random string. Used to sign/verify all session JWTs. App throws on startup if missing.
NEXT_PUBLIC_APP_URL	✔ Required	Full URL of the app (https://clearbuildusa.com). Used for email links.
RESEND_API_KEY	⚡ Recommended	Email sending. If missing, emails log to console (dev only).
STRIPE_SECRET_KEY	⚡ Recommended	Stripe server key for creating checkout sessions and customers.
STRIPE_WEBHOOK_SECRET	⚡ Recommended	Validates incoming Stripe webhooks.
STRIPE_PUBLISHABLE_KEY	⚡ Recommended	Client-facing Stripe key (NEXT_PUBLIC_).
GEMINI_API_KEY	Optional	Google Gemini API key for chatbot. Chatbot disabled gracefully if missing.

11 Security Requirements & Implementation

11.1 Authentication Security

Requirement	Implementation	Status
Password hashing	bcryptjs, 10 salt rounds	 Done
JWT signing	jose HS256, strong secret required (throws on missing)	 Done
Session cookie	HTTP-only, SameSite=Lax, 15-day expiry	 Done
Email verification	Required before login, cryptographic token via Web Crypto API	 Done
Password reset	Cryptographic token, 1-hour expiry enforced server-side	 Done
Force logout	force_logout_at column - set when team member removed; API checks on every request	 Done

11.2 Multi-Tenancy Security

Requirement	Implementation	Status
Business isolation	Every query: <code>.eq("business_id", session.businessId)</code>	 Done
Cross-business data access	Impossible - businessId comes from signed JWT, not request body	 Done
Quote approval exploit	Only "sent" quotes can be approved; status checked before update	 Done
Invoice payment hijack	Business ownership verified before Stripe checkout created	 Done

11.3 Input Validation & API Security

Requirement	Implementation	Status
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Settings field whitelist	ALLOWED_SETTINGS_FIELDS constant - only whitelisted columns can be updated	 Done
Negative amounts	Math.max(0, parseFloat(v)) enforced on all line item numeric inputs	 Done
Search query length	Max 200 chars on search endpoints, returns empty on exceeded	 Done
Role-based guards	Owner/Admin required for: payments, change order approval, notification sending	 Done
Tax calculation	Computed server-side from raw fields (not from client totals) to prevent manipulation	 Done
Booking link contact validation	contact_id verified to belong to same business before insert	 Done

11.4 Stripe Webhook Security

- Webhook signature verified using STRIPE_WEBHOOK_SECRET on every incoming event
- Only processes known event types (checkout.session.completed, customer.subscription.*)
- Idempotent handler - replaying a webhook does not double-record a payment

PRE-LAUNCH CHECKLIST

1. Confirm PLAYWRIGHT_TEST_SECRET is NOT set in Hostinger production environment (the /api/test/setup route is gated by this secret and returns 403 if it's missing - but best practice is to delete the route before launch).
2. Confirm JWT_SECRET is a strong 32+ character random string in production.
3. Confirm Stripe live keys (not test keys) are set in production.
4. Run one end-to-end Stripe payment test in production before opening to customers.

12 Testing Scenarios

12.1 Test Coverage

Test Suite	Framework	Tests	Status
Authentication flows	Playwright (E2E)	12	✓ Passing
Contact management	Playwright (E2E)	15	✓ Passing
Quote flows	Playwright (E2E)	18	✓ Passing
Invoice flows	Playwright (E2E)	14	✓ Passing
Project management	Playwright (E2E)	16	✓ Passing
Payments	Playwright (E2E)	10	✓ Passing
Navigation	Playwright (E2E)	12	✓ Passing (6 skipped - mobile only)
Total		97 / 103	✓ 94% passing

12.2 Critical Manual Test Scenarios

#	Scenario	Expected Result
1	Register → Verify email → Login	Full auth flow completes, lands on Dashboard
2	Create quote with 3 line items, send via email, customer approves	Quote status → Approved, approval logged in activity
3	Convert approved quote to invoice, mark as paid	Invoice created, payment recorded, amount_due = 0
4	Invite team member → they register → appear in team list	member appears with correct role
5	Remove team member → they try to access app	Immediately redirected to login (force_logout)
6	Business A user tries to access Business B's invoice URL	404 or empty response - no data leaked

7	Stripe checkout for Monthly plan	Subscription activated, /subscription shows active plan
8	Create scheduling slot, generate booking link, public user selects slot	Meeting created, slot marked as booked
9	Staff user tries to record a payment	Blocked by role check - 403 Forbidden
10	Create change order for a project, approve it	Change order status → Approved

12.3 Pre-Launch Smoke Test Checklist

- ☐ Landing page loads correctly on mobile and desktop
- ☐ Registration email is delivered (check Resend dashboard)
- ☐ Login works with a real production account
- ☐ Dashboard shows correct data for a business with sample records
- ☐ Stripe Monthly plan checkout completes and subscription activates
- ☐ Invoice online payment works end-to-end in production
- ☐ Quote public preview page loads without login
- ☐ Public booking page loads and slot can be selected
- ☐ All 9 settings sections save without error
- ☐ AI chatbot responds in under 5 seconds

13.1 Deployment Pipeline

Step	Action	Tooling
1	Developer commits and pushes to GitHub master branch	git push origin master
2	Hostinger detects push via GitHub webhook	Hostinger auto-deploy integration
3	Hostinger pulls repo and runs <code>npm run build</code>	Next.js build
4	If build passes, server restarts with new code	Hostinger Node.js process manager
5	Live at clearbuildusa.com	DNS: Hostinger → clearbuildusa.com

13.2 Pre-Deployment Checklist

- ☐ Run `npm run build` locally - zero TypeScript errors
- ☐ Run `npm run lint` - zero ESLint errors
- ☐ All critical Playwright tests passing (`npm run test`)
- ☐ Environment variables confirmed set on Hostinger
- ☐ Stripe live keys (not test) configured
- ☐ Test `/api/test/setup` endpoint returns 403 in production
- ☐ Verify SSL certificate active on clearbuildusa.com
- ☐ Test email delivery via Resend production key
- ☐ Confirm Stripe webhook endpoint points to `https://clearbuildusa.com/api/webhooks/stripe`

13.3 Rollback Plan

- Identify the last stable commit hash in GitHub history
- Push that commit to master: `git revert HEAD` or `git reset --hard <commit> && git push --force`
- Hostinger auto-deploys the reverted code
- Estimated rollback time: 3–5 minutes

13.4 Monitoring

What to Monitor	Tool
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Server uptime	Hostinger dashboard, or add UptimeRobot (free)
Error logs	Hostinger server logs / console.error output
Payment failures	Stripe Dashboard → Events
Email delivery	Resend Dashboard → Logs
Database health	Supabase Dashboard → Database
User activity	audit_events table in Supabase



COMPLETED - April 2026

Phase 1: Foundation

Core auth, contacts, quotes, invoices, projects, payments, team management. Custom design system. Supabase integration.



COMPLETED - May 2026

Phase 2: Full Redesign & Polish

42-screen UI redesign to professional design prototype. Change orders, scheduling, communications, templates, notifications, AI chatbot, landing page, admin panel, Playwright test suite (97/103 passing).



COMPLETED - May 2026

Phase 3: Security Audit & Bug Fixes

Full 28-bug audit. All P1 critical and P2 high bugs fixed. SQL migrations completed. Pagination implemented. Accessibility improvements.



NOW - May 2026

Phase 4: Launch Preparation

Production environment verification, Stripe live key setup, smoke testing, final documentation (this document).



TARGET - June 2026

Phase 5: Public Launch

Marketing campaign activation, first 50 paid customers goal, support team readiness.



Q3 2026

Phase 6: Growth Features

Project Gantt chart, mobile app (React Native), QuickBooks export, more integrations, referral programme.



Q4 2026

Phase 7: Scale

AI-powered estimate generation from photos, customer-facing project portal, payroll/labor hours tracking, integration marketplace.

15 Risks & Assumptions

Key Assumptions

- Target contractors use smartphones and are comfortable with web apps
- US contractors are willing to pay \$49/month for an all-in-one tool (validated by ServiceTitan at \$10K+/yr)
- Stripe is available and reliable in the US contractor market
- Email (Resend) is the primary notification channel; SMS integration may be needed later
- 15-day trial is sufficient to demonstrate value and convert
- Hostinger provides sufficient reliability and performance for early growth (upgrade to Vercel/AWS may be needed at scale)

Risk Register

Risk	Probability	Impact	Mitigation
Low trial → paid conversion	Medium	High	In-app onboarding tour, email drip campaigns, personal outreach to trial users who haven't activated key features
Stripe payment failure in production	Low	High	Test end-to-end before launch. Monitor Stripe Dashboard daily post-launch.
Hostinger downtime / performance issues	Medium	Medium	UptimeRobot monitoring. Migration plan to Vercel/AWS if needed.
Competitor launches similar product	Medium	Medium	Speed to market. Niche focus on US small contractors. Price advantage vs ServiceTitan/Jobber.
Supabase data loss or outage	Low	High	Enable Supabase automatic daily backups. Set up point-in-time recovery.
GDPR / data privacy compliance	Low (US market)	Medium	Privacy policy published. Data stored in US region. CCPA awareness for California users.
Gemini API cost at scale	Low (currently)	Low	Monitor Google Cloud billing. Add rate limiting per business if costs spike.

Product FAQ

Q: Can a contractor use Clear Build USA without a business registered?

A: No. After email verification, new users are redirected to /business-setup and must create a business before accessing any app features. This is enforced in middleware.

Q: Can one user belong to multiple businesses?

A: Yes. The business_members table supports many-to-many. The active business is in the JWT. The top bar business switcher lets users switch context without logging out.

Q: What happens when a trial expires?

A: The business enters restricted_mode. Create flows show upgrade prompts. Existing data remains accessible in read-only mode. No data is deleted.

Q: Are customer-facing quote/invoice pages secure?

A: Quote preview pages are public (by design - customers don't have accounts). They are read-only except for the approval action, which is protected: only quotes in "sent" status can be approved, preventing replay or double-approval attacks.

Q: How does online invoice payment work?

A: The contractor sends the invoice to the customer. The customer clicks "Pay \$X" on the invoice page → Stripe Checkout is created → customer pays securely on Stripe's hosted page → Stripe sends a webhook → the invoice is marked as paid automatically.

Q: Is there a mobile app?

A: The web app is fully responsive and works on mobile browsers. A dedicated React Native mobile app is on the Q3 2026 roadmap.

Q: How is my data isolated from other businesses?

A: Every database query is filtered by the businessId in your signed JWT. This is enforced server-side on every API call - no user-supplied business ID is trusted. Cross-business data access is architecturally impossible.

Q: What email provider is used?

A: Resend (resend.com). Production emails are sent from a configured domain. In local development without a Resend key, emails are logged to the console so development works without credentials.

Q: Where is data stored?

A: PostgreSQL database on Supabase (US region). File uploads (logos, etc.) stored in Supabase Storage.

Q: What is the admin panel for?

A: The admin panel (/admin) is an internal tool for the Clear Build USA team - not for customers. It provides full visibility into all businesses, users, subscriptions, and platform analytics. It has a separate login and is not linked from the main app.

